



RV College of Engineering®

Mysore Road, RV Vidyaniketan Post,
Bengaluru - 560059, Karnataka, India

Team ID	FN13			
Semester	6 th			
Course :	Interdisciplinary Project			
Team Details		Program	USN	Name
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	2	AIML	1RV24AI403	Harish Appasab Savalagi
	3	Biotechnology	1RV23BT001	A J Deeksha
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Project Title	AI-Based Multi-Label Food Spoilage Detection Using Image Analysis and IoT			
Internal Guide				
Name Designation & Department	Prof. Mekhala Vinod Purohit, Assistant Professor, Dept. of Computer Science and Engineering			

INTRODUCTION: Food spoilage leads to significant food waste and health risks, while existing systems primarily perform binary classification (fresh vs rotten) and lack integration of environmental factors. There is a need for an AI-based system that can classify fruits and vegetables into multiple spoilage stages using image analysis, along with IoT-based environmental sensing for improved accuracy and real-world applicability.

OBJECTIVES:-

1. To develop a system that **classifies fruits and vegetables into multiple spoilage stages** using deep learning models.
2. To **integrate IoT sensor data (temperature, humidity, gas levels)** with image-based analysis for improved spoilage detection.
3. To design a **user-friendly interface** that provides spoilage stage predictions along with confidence scores and visual explanations.

METHODOLOGY: A dataset of fruit and vegetable images representing different spoilage stages will be collected and preprocessed (resizing, normalization, augmentation). Deep learning models such as CNN, MobileNetV2, and ResNet50 will be trained to classify images into four spoilage stages. IoT sensors (MQ-135, DHT11) will collect environmental data like gas concentration, temperature, and humidity, which can be combined with image-based predictions. Model performance will be evaluated using accuracy, precision, recall, and F1-score, and Grad-CAM will be used for visualization. A simple web interface will allow users to upload images and view predictions.



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SOFTWARE REQUIREMENTS: Python, TensorFlow / PyTorch, OpenCV, NumPy, Matplotlib, Scikit-learn, Streamlit or Flask, Jupyter Notebook / Google Colab, Open Source Dataset.

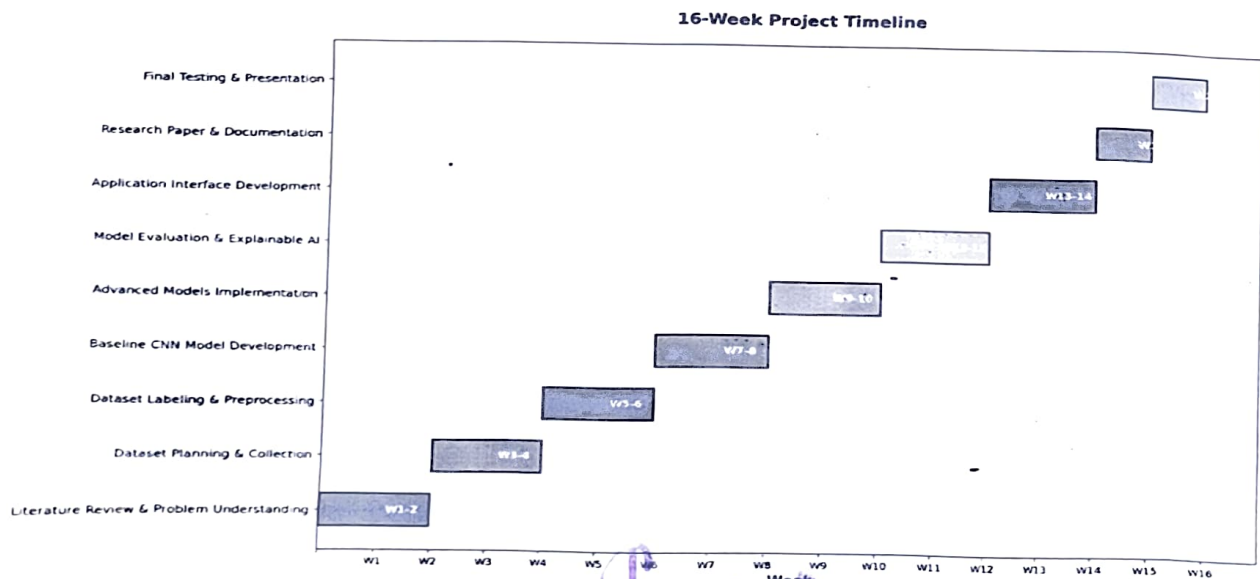
HARDWARE REQUIREMENTS: Arduino Uno / ESP32, MQ-135 Gas Sensor, DHT11 Temperature and Humidity Sensor, Buzzer, Jumper wires, Breadboard, Power supply.

INTERDISCIPLINARY RELEVANCE: The project integrates multiple disciplines:

1. Biotechnology: Identification of biological spoilage indicators such as mold growth and discoloration.
2. Artificial Intelligence / Machine Learning: Development of deep learning models for image classification.
3. Data Science: Dataset preprocessing, evaluation metrics, and performance analysis.

INNOVATION / CONTRIBUTION TO THE FIELD: Most existing food spoilage detection systems perform binary classification (fresh vs rotten). This project introduces multi-stage spoilage classification, providing more detailed information about the deterioration level of food items. The system also integrates Explainable AI (Grad-CAM), IoT-based hardware sensors (MQ-135, DHT11), and a user-friendly web interface, demonstrating practical AI applications for food quality monitoring and food waste reduction.

TIMELINE (GANTT CHART):



18/03
Guide

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